

Simulations

Thursday, July 10, 2025 2:31 PM

Goal: Match the simulation results for each stage with the experimental traces generated to verify if it agrees with the predicted results.

1. Measured the values of each component for a given channel from each stage filter board.
2. Updated the values on the circuit

For reasonable values: measure @ C- 1kHz, L 100kHz

Questions:

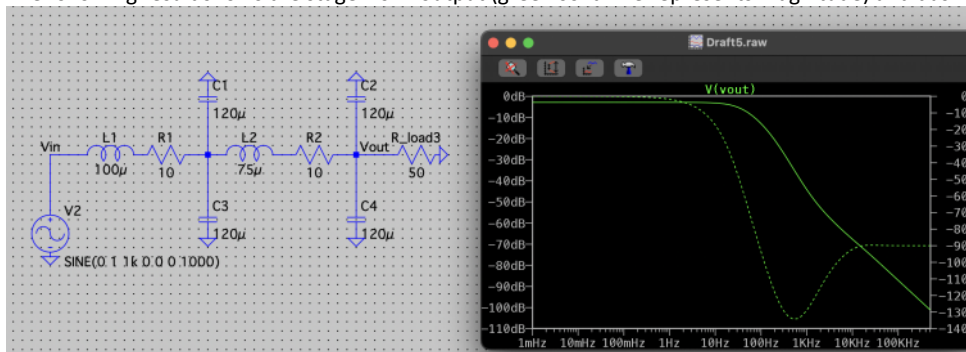
1. Some values don't match with the ones in the document.

- 1st Stage:

- Capacitance was found to be 120uF instead of 100uF (doc) @1kHz. This was measured using a LCR Meter at 1kHz and 100kHz both. For 100kHz it gave a value around -100nF which doesn't make sense.
- Resistors matched with the document (10 ohms).
- The first inductor was measured to be around 100uH and the second one 74uH (100uH in doc) @1kHz. Similarly 95.2uH and 5uH @100kHz which is very off.

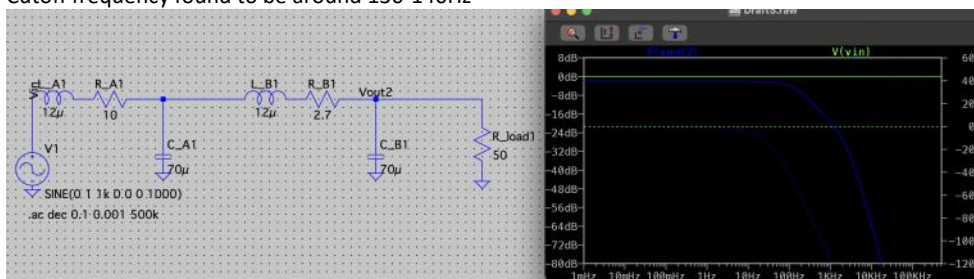
Cutoff frequency found to be around 41Hz

The following result shows the stage 1 sim output (green solid line represents magnitude) and dashed line represents phase.



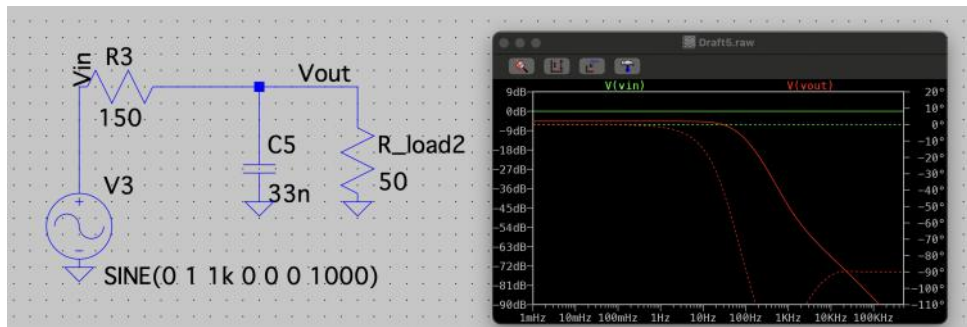
- 2nd Stage:

- Capacitance was found to be 70uF @1kHz. For 100kHz it again gave a value around -100nF (-95ish).
- First resistor was matching (10 ohms) but the second one didn't match 2.7ohm (in doc it's 10 ohms).
- The inductors were found to be around 10uH (0L @1kHz) and 12uH (@100kHz) which again didn't match with the doc.
- Cutoff frequency found to be around 130-140Hz



- 3rd Stage:

- This was perfectly fine. It matched with all of the values in the doc.
- For the following curve, we can see that the cutoff freq (-3db) happens at around 40hz



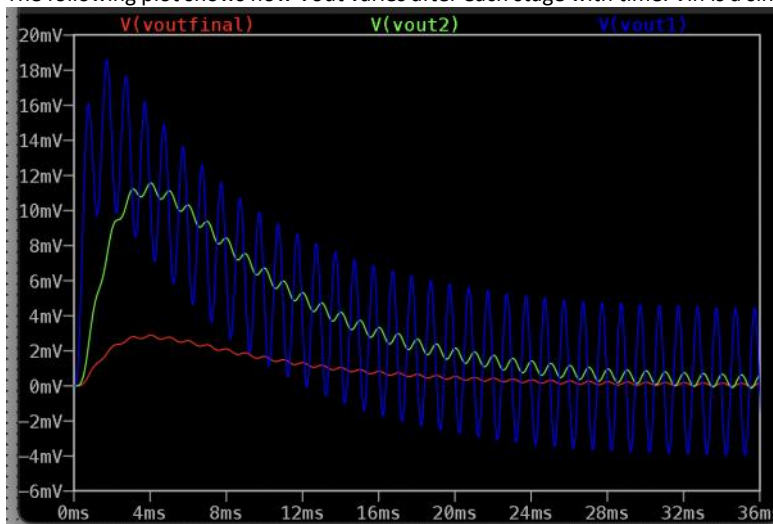
What should be the next steps based on the above simulation plots?

1. Changing the R_{load} to a higher value or the third stage resistor to a lower value I'm getting better results with initial attenuation close to 0db for low frequencies due to voltage divider rule.
2. Some components' values don't match with the document.
3. When using LCR meter to measure sometimes based on the frequency (1kHz or 100kHz) it keeps fluctuating and not a clear answer depending on the point of contact for taking the measurement.
4. I don't think the cutoff frequencies make sense right now for the above stages just based off the curves. But after generating the plots with oscilloscopes and everything if it matches these predicted simulations it should verify the correct behavior of these low pass filter stages.
5. When measuring the capacitances of capacitors in the stage 1 we getting 120uF. It should be 100uF. Is it because of 2 parallel capacitors in parallel? Or is it because of some other possible configuration. Confirm w Aman.

Measurements using the signal generator and a spectrum analyzer:

1. Programming both of them so that they are measuring at the same time (M)
2. Is it okay if we just see the response after 9kHz as the spectrum analyzer and the signal generator have this range or do we have to check for <1kHz responses. This should also be able to verify our sim plots generated in LTSpice.

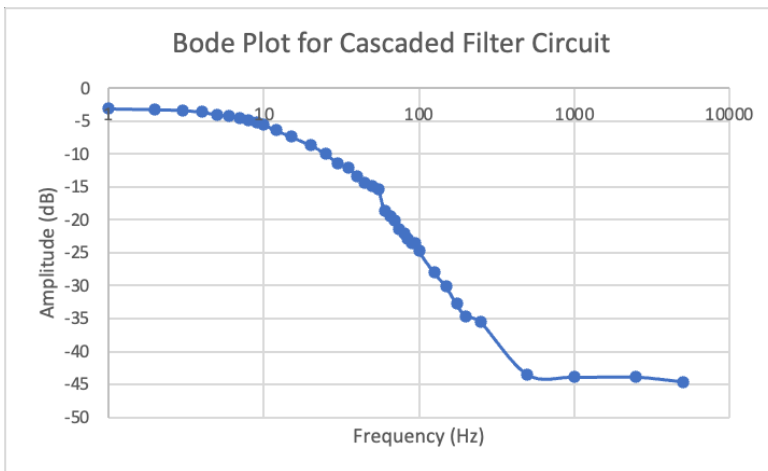
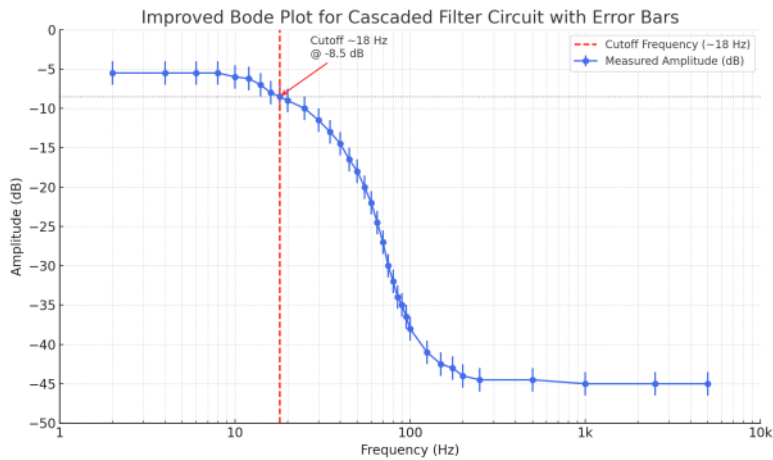
The following plot shows how V_{out} varies after each stage with time. V_{in} is a sinusoidal with 1V amplitude and 1kHz frequency :



Based on the simulations cutoff frequency for the cascaded stages circuit was found to be around 5.496 Hz.

Measurements:

Feeding in an input sine signal of frequency from 20 Hz to 5000 Hz through all the stages of the circuit we found the bode plot roughly looks like the following: Cutoff Frequency $\sim 20\text{Hz}$



Some questions:

1. When taking the measurements, for higher frequencies I noticed that the input frequency from the signal generator and the measured frequency by the oscilloscope were very different. Do we take the avg f shown on the oscilloscope or the input f?
2. Couldn't do it precisely for higher frequencies (there was a lot of noise)
3. The above plots make some sense as they show the attenuation within a reasonable range of frequencies.
- 4.

Match the above results with the theoretical model.

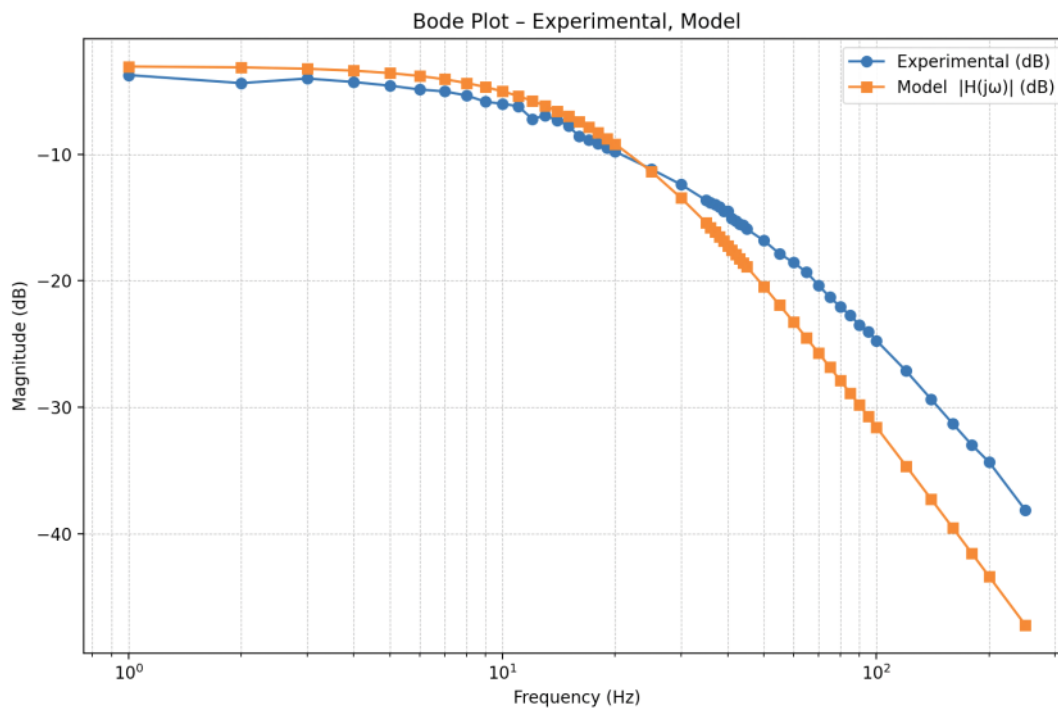
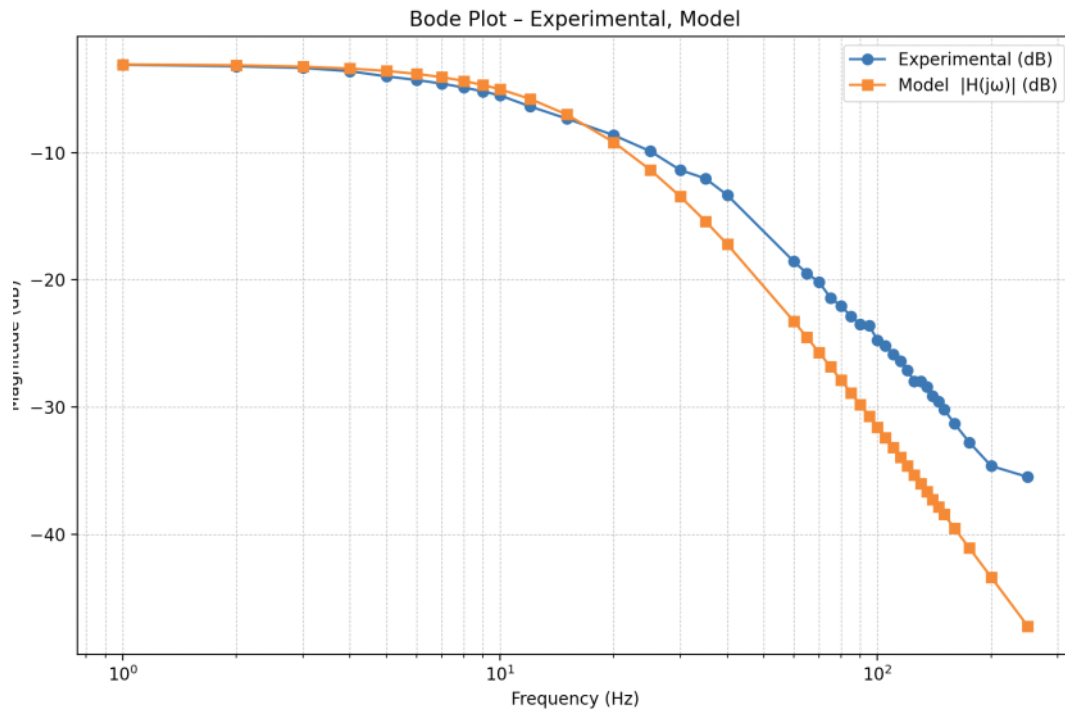
$|H(w)| = \text{nth polynomial} / \text{mth polynomial sums with maybe } n, m = (1, 2)$

Plot this theoretical on top of the above graph

See if they match then proceed further

Make the measurements again individually for each stage of the fc

- Took measurements twice, the following pics show the curves obtained after optimizing the parameter values using the scikit(python)
- The first one shows for the data collected earlier and the second one is the latest (within ~3 db when compared to the first one) and much smoother when compared to the first one.
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Current Values of the components:

Input Impedance – 50 ohm

Output – 50 ohm

Observations:

Changing L to 12u from 0.74u and 0.47u makes the curve more steeper

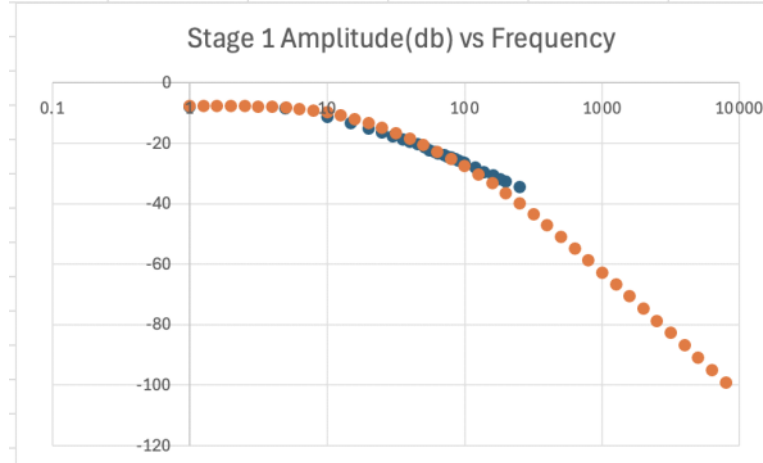
Capacitors to 2.2u from 100u makes more steeper and less initial attenuation

- -10db, -95db at 10khz, 100khz respectively

Individual Stages measurements

=> Stage 1 behavior perfectly matches with the LTSpice simulation

- Both start at around -8db and follow a similar trend when amplitudes compared at different frequencies
- Near higher frequencies (like 250ish) there is a slight difference of ~4 dB between the two.



=> Stage 2 experimental results were very off for some reason. I'm not sure if it was because of the noise or what but I tried measuring it twice. The second time it was somewhat reasonable but then the cutoff frequency range from the plot seemed above 200kHz which was not available on the LF channel.

So, after switching to the RF channel, I was trying to take some readings today but it was not attenuating the signal. Vin and Vout were roughly the same even after going beyond 200kHz. Also, showed Aman but we couldn't really figure out anything. I'll try doing the measurements once last time again for the second stage and see if it makes any sense.

One possible explanation could be that there's a 50 ohm input impedance on the RF channel but it isn't mentioned on the LF so maybe that could be the reason.

=> Stage 3 results were also somewhat different. In sims, it starts at around -14db whereas in the exp it starts at roughly -5.5db. It's possible that this could just be an offset so if the slope matches then it should be fine.

-> Sims; 11, 50, 150 khz

-> Changing Rload in 3rd stage sim to 5000 from 50 results in a more accurate curve that matches the experimental observations (within ~2db at higher freq like 150khz)

Updated FC

